

Chapter 21. Extracorporeal Carbon Dioxide Removal

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Artificial Organs for Respiratory Failure

Extracorporeal carbon dioxide removal (ECCO₂R) refers to a technique of life support focused on the removal of CO₂ from blood rather than improving blood oxygenation.¹ This chapter introduces the concept of ECCO₂R and discusses some of the experience and future on this exciting topic. ECCO₂R has been developed mainly with a view to applying it in patients with the most severe form of acute respiratory distress (ARDS).² We will try, however, to widen the perspective to cover also the possible role of ECCO₂R in the prevention of hyaline membrane disease³ and in the treatment of severe asthma,^{4–8} multiple bronchopleural fistulas,^{9–10} and severe chronic obstructive pulmonary disease.¹¹

The mainstay of supportive treatment in ARDS is mechanical ventilation, a lifesaving procedure introduced in the management of patients with bulbar polio in the great epidemic that struck Copenhagen in 1952. These patients, paralyzed by polio, required long-term artificial ventilation;¹² they became the first critical care patients. The use of mechanical ventilators later was extended to all patients with severe acute respiratory failure, whose main problem often was altered gas exchange and not respiratory muscle weakness or paralysis. The critical care profession witnessed both the pros and cons of optimizing gas exchange through use of the ventilator: The focus shifted from high to low tidal volumes, from high to low airway pressures, and from high to lower inspired oxygen fractions (F_IO₂). It is provocative to consider how we support the failing lung. In ARDS, we use an artificial organ (the ventilator), which is designed to substitute for the respiratory muscles rather than to act as a gas exchanger. Technology has been the limiting factor for a widespread application of artificial gas exchange,¹³ but research and development continues at a promising pace.

Membrane Oxygenators, Membrane Gas Exchange, and Membrane Lungs

Extracorporeal oxygenation was first provided as a heart–lung machine to render major cardiovascular surgery feasible and safe.¹⁴ The first oxygenators were based on bubbling of oxygen through the blood or filming of blood in an oxygen atmosphere. To avoid the problems caused by the direct contact between blood and gas,^{15,16} Kolff designed a membrane oxygenator,¹⁷ which Clowes¹⁸ and Kolobow¹⁹ developed further into clinically applicable membrane gas exchangers.^{20–22} Attention was focused on extracorporeal oxygenation, and the term *extracorporeal membrane oxygenation* (ECMO) was coined. Very little attention was paid to concurrent CO₂ removal: Hypocapnia was recognized as a common annoyance to be prevented by adding CO₂ to the gas ventilating the oxygenator.

Pioneers

In 1967, Ashbaugh et al²³ described ARDS. Soon this became a very common diagnosis in critical care, and mechanical ventilation moved to center stage as the main supportive therapy.²⁴ To optimize oxygenation, tidal volumes of 10 to 15 mL/kg were recommended, levels of positive end-expiratory pressure (PEEP) ranged from 5 to 60 cm H₂O.^{25,26} With these ventilator settings, patients with ARDS experienced high inspiratory pressures and gross disruption of lung parenchyma. Barotrauma was common.^{27–29} In 1972, Hill et al³⁰ published the first successful ECMO application in a patient with ARDS. Over the next few years, 217 patients with acute respiratory failure were supported with ECMO.³¹ In 1973, a group of pioneers initiated a National Institutes of Health (NIH)-sponsored randomized trial of ECMO in severe ARDS. The entry criteria (Table 21-1) were intended to enroll a population with a 70% mortality rate. In fact, final mortality was 90% in both the control and ECMO groups.³² The NIH-ECMO study proved that long-term extracorporeal life support was feasible, but it did not show any benefit on survival. Consequently, adult ECMO almost stopped,³³ with the exception of a few centers, such as the one where Bartlett conducted innovative studies on extracorporeal life support in infants and adults.

Table 21-1: Extracorporeal Membrane Oxygenation Entry Criteria: $\text{PaO}_2 \leq 50$ mm Hg (Repeated Three Times)

Study Entry Speed	Entry Testing Period (hours)	FI_{O_2}	PEEP, cm H ₂ O	Qs/Qt	PaCO_2 mm Hg	ICU Care Duration Before Entry Testing (hours)
Rapid	2	1.0	≥ 5	—	30 to 45	—
Slow	12	≥ 0.6	≥ 5	≥ 0.3	30 to 45	≥ 48
ECMO EXCLUSIONS						
Contraindication to anticoagulation (e.g., gastrointestinal bleeding, recent cerebrovascular accident, chronic bleeding disorder) Pulmonary artery wedge pressure >25 mm Hg Mechanical ventilation >21 days Severe chronic systemic disease or another clinical condition that in itself greatly limits survival, for example, Irreversible central nervous system disease Severe chronic pulmonary disease ($\text{FEV}_1 < 1$ L, $\text{FEV}_1/\text{FVC} < 30\%$ of predicted, chronic $\text{PaCO}_2 > 45$ mm Hg, chest X-ray evidence of overinflation or interstitial infiltration, previous hospitalization for chronic respiratory insufficiency) Total body surface burns $>40\%$ Rapidly fatal malignancy Chronic left-ventricular failure Chronic renal failure Chronic liver failure						

FEV_1 , forced expiratory volume in 1 s; FVC , forced vital capacity; Qs/Qt , right-to-left shunt fraction.

Source: Data from Zapol et al³² and the National Heart, Lung, and Blood Institute.⁸⁵

Meanwhile, astute observers pondered the question: Why did the NIH-ECMO study fail?³⁴

Why Did the National Institutes of Health Extracorporeal Membrane Oxygenation Study Fail?

ECMO was aimed at buying time for the lung to rest and heal.³⁵ This could not be achieved under the persisting damage caused by high tidal volumes and pressures, however. Lung management in the ECMO group was not much different from that in the control group; this fact could explain the similarities in survival.³⁴ The choice of a venoarterial bypass, aimed in part at lowering pulmonary blood flow and pulmonary artery pressure, might have contributed to severe maldistribution of pulmonary blood flow. In turn, this might have deprived the lungs of enough blood flow to defend against the damage of high-tidal-volume, high-pressure ventilation.³⁶

The Concept of Extracorporeal Carbon Dioxide Removal

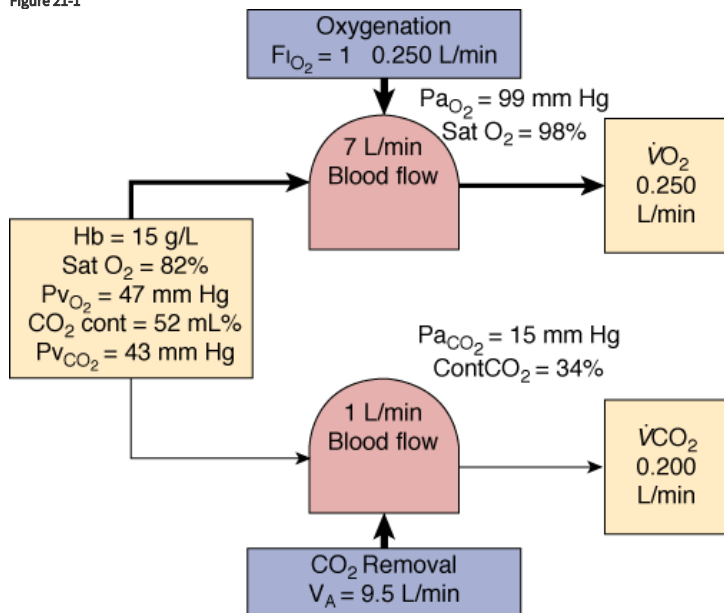
In 1976, Kolobow et al¹ noted that membrane oxygenators more appropriately constituted membrane lungs than just oxygenators. They observed that a membrane lung can exchange CO_2 much more easily than it can oxygenate blood. They explored the potential of ECCO₂R in a series of innovative experiments.

Dissociating Respiratory Functions

Blood oxygenation and CO_2 removal take place through different mechanisms.³⁷ When normal venous blood reaches the lungs, its mixed venous oxygen tension (PvO_2) is typically 47 mm Hg, and mixed venous carbon dioxide tension (PvCO_2) is 43 mm Hg (Fig. 21-1). Let us assume that oxygen consumption and CO_2 production amount to 250 and 200 mL/min, respectively. The hemoglobin carried in venous blood (150 g/L) is normally 70% to 85% saturated; the lungs therefore can add just 40 to 60 mL of oxygen per liter of venous blood. Thus, to fulfill the requirements for oxygenation, we need a blood flow of at least 4 to 7 L/min. Note that ventilation is not required to oxygenate blood, whereas what is strictly needed is enough oxygen to compensate for its consumption and maintain a constant alveolar concentration. In summary, oxygenation requires a high blood flow (4 to 7 L/min) and a small supply of

oxygen (250 mL/min). The opposite applies to CO₂ removal. PvCO₂ content is at least double the maximum O₂ content (Fig. 21-2). Consequently, the normal CO₂ production per minute can be removed easily from less than 1 L of blood, provided that ventilation is high enough.

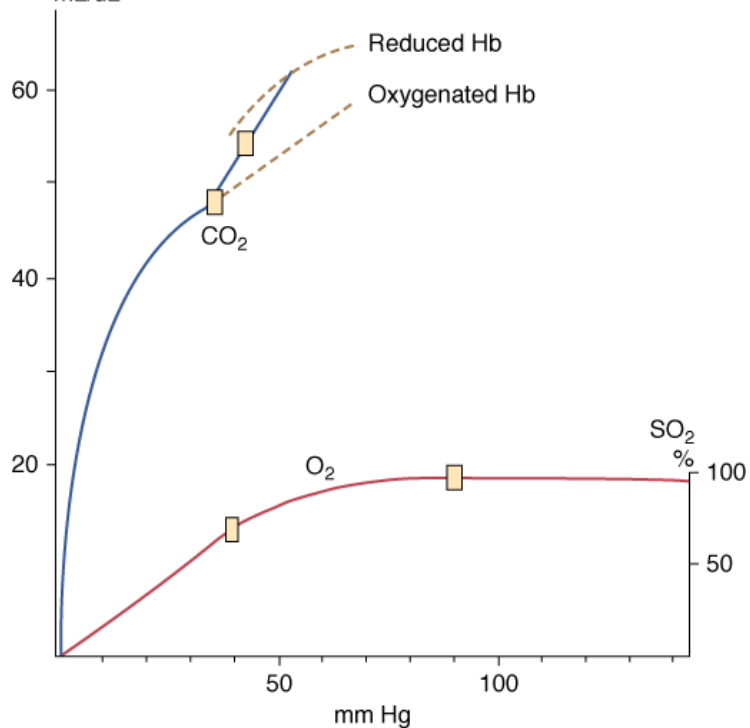
Figure 21-1



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Dissociation of oxygenation and CO₂ removal. For normal mixed venous blood (v), oxygenation requires normal pulmonary blood flow and a continuous O₂ supply equal to O₂ consumption (250 mL/min in this example) without any ventilation. Removal of CO₂ can be accomplished by a reduced pulmonary blood flow if this is matched by a sufficiently high alveolar ventilation. (Reproduced, with permission, from Gattinoni L, Pesenti A, Kolobow T, Damia G. A new look at therapy of the adult respiratory distress syndrome: motionless lungs. *Int Anesthesiol Clin*. 1983;21:97–117.)

Figure 21-2
mL/dL

Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

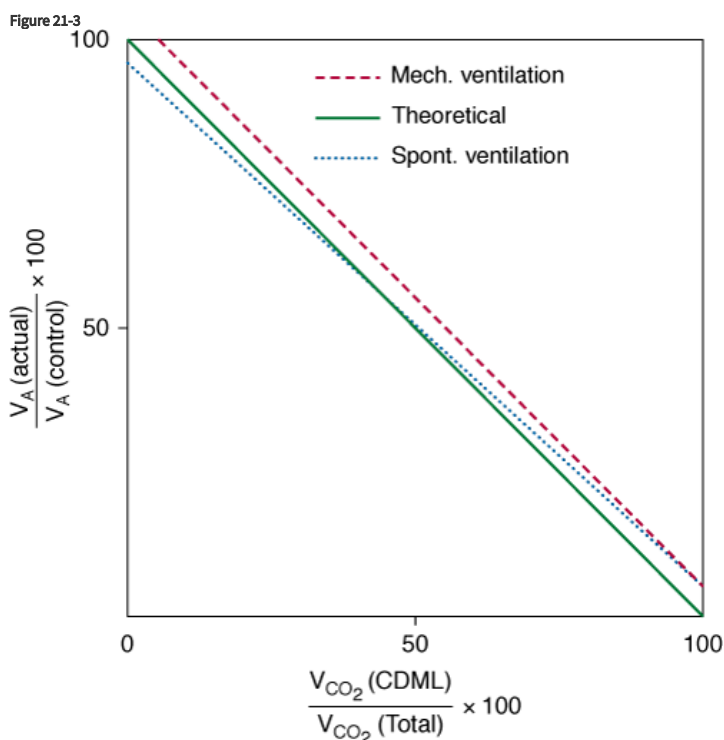
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Gas dissociation curves for O₂ and CO₂ in blood. The *solid rectangles* represent arterial and mixed venous blood points. The line connecting the CO₂ dissociation curves for reduced and oxygenated hemoglobin (Hb) represents the nominal in vivo CO₂ dissociation curve as mixed venous blood becomes oxygenated in the lung (Haldane effect).

In conclusion, oxygenation requires a high blood flow, whereas CO₂ removal can be achieved at low blood flows. Kolobow et al³⁸ exploited the concept that if CO₂ is removed by a membrane lung through a low-flow, high-ventilation venovenous bypass, then oxygenation can be maintained by the natural lung without any ventilatory constraint. Physicians now had an opportunity to adjust the ventilator settings free of the constraints of tidal ventilation.

Extracorporeal Carbon Dioxide Removal and the Control of Spontaneous Breathing

When CO₂ was removed by the membrane lung, unsedated lambs decreased their spontaneous ventilation, reducing both respiratory frequency and tidal volume.³⁹ Changes in alveolar ventilation were highly predictable, targeted toward a constant pH and PaCO₂. The greater the removal of CO₂ by the membrane lung, the greater was the decrease in alveolar ventilation (Fig. 21-3). The amount of CO₂ excreted by the natural lung decreased to maintain the total CO₂ removal (sum of membrane lung plus natural lung) constant and equal to the rate of CO₂ production by the body.



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Alveolar ventilation (percent of control values) as a function of extracorporeal CO₂ removal (percent of total CO₂ production, i.e., V_{CO_2} of carbon dioxide membrane lung [CDML] plus natural lung). The theoretical values are computed assuming that PaCO₂ and total V_{CO_2} are constant throughout the procedure. (Reproduced, with permission, from Gattinoni L, Pesenti A, Kolobow T, Damia G. A new look at therapy of the adult respiratory distress syndrome: motionless lungs. *Int Anesthesiol Clin*. 1983;21:97-117.)

We can postulate that the control of spontaneous breathing in chronic obstructive pulmonary disease patients, while resetting the CO₂ balance with extracorporeal removal. Could be easy to achieve and predictable. We have experience about the control of respiratory drive in patients with ARDS during their recovery phase, when extracorporeal CO₂ removal can be titrated to the needs of the patient until weaning is completed.⁴⁰ Unfortunately, very little is known about the impact of ECCO₂R upon the respiratory drive in the acute phase of ARDS.

Apneic Oxygenation

When CO₂ production is entirely removed by a membrane lung, then ventilation is no longer needed. The lung can be kept motionless, provided the alveolar oxygen concentration is kept constant by continuously supplying oxygen to match the body's O₂ consumption.³⁸ This process, known as *apneic oxygenation*,⁴¹ is otherwise normally limited by the CO₂ rise. In this case, the membrane lung avoids an increase in CO₂. Apneic oxygenation therefore can be maintained at will by ECCO₂R: The lung can be kept completely motionless, the ultimate goal being lung rest and recovery.

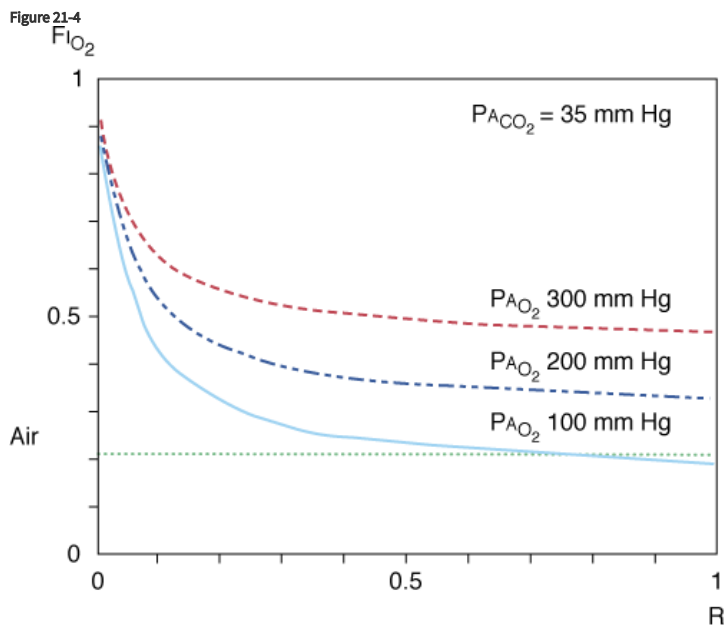
Low-Frequency Positive-Pressure Ventilation

A decrease in respiratory compliance and functional residual capacity was noticed during apneic oxygenation. This led to the introduction of low-frequency positive-pressure ventilation (LFPPV). LFPPV with ECCO₂R⁴² is simply apneic oxygenation to which a few (deep) breaths (sighs) per minute are added. With LFPPV-ECCO₂R, respiratory compliance and functional residual capacity can be maintained at baseline for days or weeks.

Alveolar Partial Pressure of Oxygen Control during ECCO₂R

During ECCO₂R, the respiratory quotient (R), that is, the ratio of CO₂ removal from the natural lung to **oxygen** uptake, changes according to the amount of CO₂ removed by the membrane lung. If baseline R equals 1, when we remove 50% of the body's CO₂ production by ECCO₂R, the new R of the natural lung will be 0.5. This affects alveolar **oxygen** tension (PaO₂), which follows Riley's alveolar gas equation: At any given FiO₂, when R decreases, PaO₂ decreases.

Figure 21-4 shows the changes in FiO₂ required to maintain a constant PaO₂ at varying R values. Note that PaCO₂ is held constant through the effect of ECCO₂R.³⁷



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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FiO₂ required to maintain a PAO₂ of 100, 200, and 300 mm Hg at constant PaCO₂ as function of pulmonary respiratory quotient (R) according to Riley's alveolar air equation. (Reproduced, with permission, from Gattinoni L, Pesenti A, Kolobow T, Damia G. A new look at therapy of the adult respiratory distress syndrome: motionless lungs. *Int Anesthesiol Clin*. 1983;21:97-117.)

When ECCO₂R equals 100% of the CO₂ produced by the body, then alveolar ventilation is nil, and FiO₂ must be raised to 1 to compensate for **oxygen** consumption. This does not at all mean that the alveolar **oxygen** concentration must be 100%. At this extreme of physiology, the gas mixture ventilating the membrane lung is of the utmost importance. At a steady state and during apneic oxygenation, the only gas exchange taking place in the lung is **oxygen** consumption. No nitrogen, CO₂, or water exits or enters the alveolar gas. If the gas ventilating the membrane lung is 100% humidified **oxygen**, then alveolar gas will be 100% **oxygen** minus CO₂ and water. If nitrogen (N₂) is added to the gas ventilating the membrane lung, then mixed venous blood P_{N₂} will equilibrate with P_{N₂} of the gas in the membrane lung, causing a corresponding change in PaO₂.³⁸ During apneic oxygenation, despite the need to keep FiO₂ at 1, PaO₂ will be determined by the P_{N₂} of gas in the membrane lung. The role of membrane lung P_{N₂} in preventing reabsorption atelectasis during ECCO₂R is entirely speculative but based on accepted physiology.⁴³

Because of the Haldane effect, PaCO₂ (and PaCO₂) will be higher than PvCO₂ secondary to changes in the CO₂ dissociation curve related to the oxygenation of hemoglobin. During LFPPV-ECCO₂R, PaO₂ is mostly regulated by FiO₂. If a continuous **oxygen** flow is added to compensate for **oxygen** consumption taking place between breaths, then many factors come into play. Experimental data exemplify the complexity of these mechanisms.⁴⁴

Rationale for Extracorporeal Carbon Dioxide Removal Use in Acute Respiratory Distress Syndrome

The goals of ECCO₂R are different from those of the 1974 to 1977 ECMO trial³² (Table 21-2). The reasons for these differences are rooted in ARDS pathophysiology. As already outlined, mechanical ventilation was developed by many workers^{45–52} who pointed out how positive pressure and tidal volume interact in optimizing oxygenation. At the same time, however, the dangers and drawbacks of mechanical ventilation became apparent.²⁷ This was no surprise. Teplitz, a U.S. Army pathologist during the Vietnam War, described ARDS as “an end-stage pathologic picture which... is not a new disease process... but a result of iatrogenic modification of the pathology of noncardiogenic pulmonary edema.” This view underlined the interaction between the original insult and the evolution and damages caused by treatment.⁵³

Table 21-2: Natural (Patient) Lung Treatment and Goals: ECMO versus LFPPV-ECCO₂R

	ECMO ³²	LFPPV-ECCO ₂ R
GOALS		
Ventilation	Minimize FI _{O2} , TRADITIONAL V _T	Minimize FI _{O2} , LUNG REST
Extracorporeal circulation	Arterial Oxygenation	CO ₂ removal (to rest the lung)
TREATMENT		
Lung ventilation	V _T = 0.6 L	V _T low
	P _{PEAK} = 50 cm H ₂ O	P _{PEAK} = 35 to 40 cm H ₂ O
	PEEP = 10 cm H ₂ O	PEEP = 17 cm H ₂ O
	VR = 15/min	VR = 2 to 4/min
Lung perfusion	Low (0.1 Q _t)	High (all Q _t)

P_{PEAK}, peak airway pressure; Q_t, cardiac output; VR, ventilator rate; V_T, tidal volume.

Pontoppidan⁵⁰ suggested a tidal volume of 12 to 15 mL/kg as ideal for patients with ARDS. He also issued warnings^{27–29} about the side effects of continuous positive-pressure ventilation and the appearance of lung damage related to high, high pressure, and high volumes.

In the late 1970s and early 1980s, the deleterious effects of mechanical ventilation were elucidated, and the concept of barotrauma evolved to that of volutrauma.^{54,55} A third mechanism of damage was proposed later: the release of inflammatory mediators from the ventilated lung, leading eventually to multiple-organ failure and possibly death.^{56–60} The term *ventilator-induced lung injury* became popular. In the meantime, a revolutionary idea was gathering momentum: Why strive to maintain a normal PaCO₂ in patients with ARDS? Trying to achieve this goal can damage the lung severely, whereas accepting a higher than normal PaCO₂ may induce only minor side effects. In 1990, Hickling et al⁶¹ reported a better outcome by lowering tidal ventilation and tolerating high PaCO₂ levels. They proposed the term *permissive hypercapnia*, indicating the price to be paid for limiting barotrauma. The targets of gas exchange in ARDS changed quickly.⁶² A similar approach had been suggested previously for severe asthma.⁶³

Several studies were performed to investigate the effects of lung-protective (low-pressure, low-tidal-volume) ventilation.^{64–69} Only two studies^{64,66} demonstrated benefit, but their effect has been striking. For the first time, the mode of ventilation was shown to affect outcome. One limitation with lung-protective ventilation, however, is hypercapnia. In selected patients, low-flow ECCO₂R offers a very powerful tool for overcoming these limitations while offering total lung rest.

Hypoxemia, however, is the main characteristic of ARDS. Supportive therapy in ARDS is directed to its relief because hypoxemia is a major determinant of organ dysfunction and even may be the direct cause of death. Mechanical ventilation can be tailored to correct hypoxemia primarily by increasing FI_{O2} and airway pressures.⁷⁰ Use of high airway pressure, however, is limited by several factors mainly related to hemodynamics and barotrauma. The solution is often to decrease tidal volume and increase frequency; this solution culminates in the use of high-frequency ventilation or oscillation. With high-frequency oscillation, the mean airway pressure is in principle much higher than during positive-pressure ventilation. High-frequency ventilation

has proven safe and effective in the treatment of adult patients with ARDS.^{71–73} Use of high airway pressures, however, combined with inspiratory pressure limitation may lead to insufficient carbon dioxide removal.

In summary, the major aims of ECCO₂R are to prevent ventilator-induced lung injury and barotrauma by limiting the ventilation in a nonhomogeneous lung, to put the lung to rest, and to foster healing while maintaining a selected PaCO₂. In addition, ECCO₂R enables the application of a constant mean airway pressure targeted to optimal oxygenation and free of the constraints dictated by tidal ventilation.

Clinical Applications: Venovenous Bypass

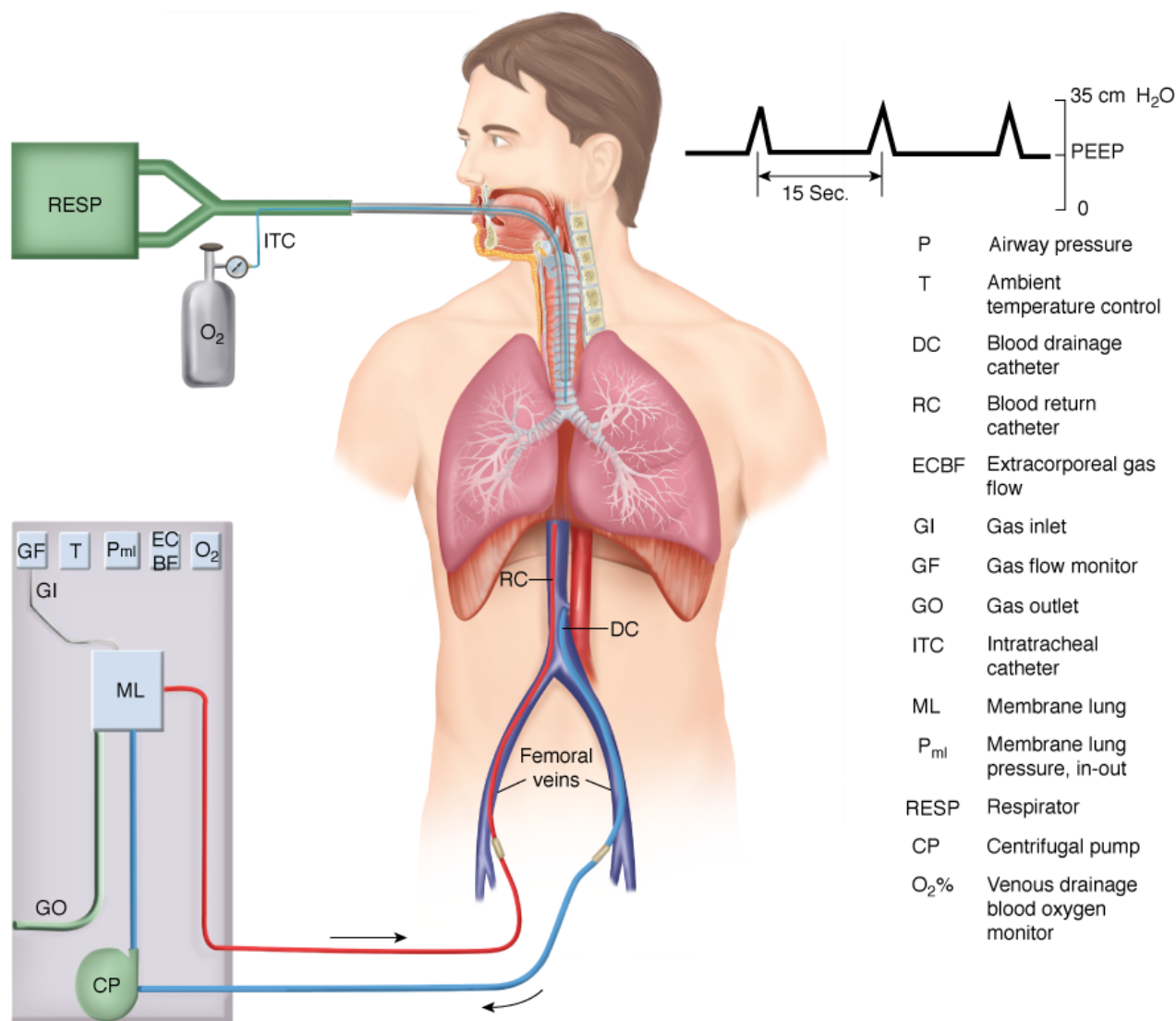
To the present time, the main indication for ECCO₂R is severe ARDS secondary to a potentially reversible cause.^{74,75} Generally accepted contraindications to extracorporeal life support and ECCO₂R include the presence of significant bleeding, surgery in the preceding 72 hours, severe brain damage, uncontrolled severe sepsis, unresolved malignancies, severe chronic systemic disease, and ARDS of a known irreversible origin.

Bypass Technique

In 1979, Gattinoni et al² reported the use of LFPPV-ECCO₂R in an adult patient with ARDS. The technique involved venovenous bypass: The common femoral and jugular veins were cannulated both distally and centrally through surgical cutdowns. The wounds and the multiple cannulation involved continuous oozing of blood and limitations in nursing care and patient mobility.⁷⁶ Subsequently, we developed a double-lumen cannulation of the femoral vein that allowed a single cutdown.⁷⁷ With saphenosaphenous bypass,⁷⁸ surgery became very superficial and distal drainage was unnecessary. The number of cannulas therefore was reduced to two. The most significant improvement in cannulation, however, came with the springwire-reinforced percutaneous cannulas,^{9,79} which are placed by a modified Seldinger technique,⁸⁰ with a shorter procedure time, practically no bleeding, a reduced risk of cannulation-site infection, and very simple decannulation.

Blood flow is normally kept at 15% to 30% of the patient's cardiac output. The system must have a capability of running at 50 to 60 mL/kg/min should we need to substitute for the natural lung oxygenation (Fig. 21-5).

Figure 21-5



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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LFPPV-ECCO₂R circuit. (Reproduced, with permission, from Gattinoni, et al. *JAMA*. 1986;256:881-886,. Copyright © 1986 American Medical Association. All rights reserved.)

Our current standard includes the use of two springwire-reinforced percutaneous femoral cannulas (20F to 28F), a centrifugal pump, and a plasma tight hollow-fiber polymethylpentene oxygenator. The entire circuit is surface heparinized to minimize the need for systemic anticoagulation.⁸¹⁻⁸³

Anticoagulation

Following an initial 50 to 100 IU/kg intravenous heparin bolus at the time of cannulation, heparin infusion is started, aiming at the selected activated clotting time (150 to 200 seconds in the case of Jostra Bioline surfaces). Surface-heparinized circuits can even be run without any systemic anticoagulation⁸⁴ for at least 12 to 48 hours as needed to stop or prevent incidental bleeding. Antithrombin III activity is maintained around 100% to promote surface-bonded and intravenous heparin function. Platelets are transfused when lower than 50,000/ μ L. When heparinized surfaces are not in use, activated clotting time and/or partial thromboplastin times of 1.5 to 2 times normal must be maintained at all times.

Clinical Management

ECCO₂R normally is started in a sedated, paralyzed patient. After the initial adjustments (which normally take 1 to 2 hours), the ventilator is set to provide a low-frequency sigh over a baseline constant PEEP (e.g., using intermittent mandatory ventilation or biphasic positive airway pressure). PEEP is adjusted to maintain mean airway pressure at the prebypass level and to prevent acute worsening of lung edema. A catheter inserted into the inspiratory line provides a constant oxygen supply and constant PEEP during the long expiratory pause. As soon as possible, attempts are made to reestablish spontaneous respiratory activity, most often in the form of pressure supported breathing with an intermittent sigh (e.g., biphasic positive airway pressure plus assisted breathing or intermittent mandatory ventilation plus pressure support).

Hemodynamics are not affected by the venovenous bypass. Changes in lung function can be followed with venous admixture measurements. Very high values of mixed venous saturation and arterial O₂ saturation suggest decreased cardiac output with an increased proportion of extracorporeal blood flow/total cardiac output. When lung function improves, weaning is attempted by decreasing and PEEP and by decreasing the CO₂ removal from the membrane lung.⁴⁰ When necessary, the extracorporeal circuit setup allows low-flow venovenous ECCO₂R to be converted into high-flow venovenous ECMO, and management then is focused on achieving a viable oxygenation despite an extremely reduced or even absent natural-lung oxygen transfer.

Complications of Venovenous ECCO₂R

We have never stopped ECCO₂R because of a technical accident. From day to day, however, various changes of circuit elements may be required, mainly involving the membrane lung and/or the centrifugal pump(s).

Bleeding always has been the major complication with extracorporeal life-support techniques. The NIH-ECMO study reported an average blood-product transfusion of 3575 mL/day.⁸⁵ In 1986, we reported⁸⁶ an average blood-product requirement of 1.8 L/day in our first forty-three patients. The use of percutaneous cannulation coupled with heparinized surfaces decreased the packed-red-cell requirement to 200 to 300 mL/day.⁷⁵ Bleeding from the chest drainage tubes is a major complication that often demands a surgical approach.⁸⁷ The major threat posed by extracorporeal support remains intracranial hemorrhage. We reported that a prebypass PaCO₂ of greater than 75 mm Hg, disseminated intravascular coagulation, or a positive brain computed tomographic scan before bypass increase the risk of fatal intracranial hemorrhage during bypass.⁷⁵

Clinical Results in Acute Respiratory Distress Syndrome Patients

It is difficult to isolate from the literature the experience with venovenous ECCO₂R because of undefined boundaries between venovenous ECCO₂R, venovenous ECMO, and partial extracorporeal CO₂ removal (PECO₂R).⁴⁰ ECCO₂R refers more to a way of managing the diseased lung rather than to technical details, and the flexible handling of the circuitry is one of the advantages of ECCO₂R. The technique can shuttle back and forth between venovenous ECCO₂R, venovenous ECMO, and PECO₂R in the same patient.

Table 21-3 is an attempt to collect the available experience on ECCO₂R, defined as such by authors, including our own data (see also Table 21-4). At the time of writing, patients with severe ARDS are still starting on ECCO₂R later and later in their illness. Whether this is wise, we have our doubts. Moreover, the time spent on bypass gets increasingly longer (139 days for our longest survival run), probably indicating increases in unmeasured elements of disease severity at the time of connection. Despite these considerations, many patients with ARDS suffer from complications related to ventilator-induced lung injury and are doomed to an unfavorable outcome. ARDS carries a substantial mortality, 31% to 39.8% in the ARDS Network studies.⁶⁶

Table 21-3: Venovenous ECCO₂R for ARDS: International Experience

Author (Ref.)	Year	Center	No. Patients	Survivors	% Survival
Wagner (87)	1990	Marburg (Germany)	76	38	50%
Bindslev (82)	1991	Stockholm (Sweden)	14	6	43%
Brunet (121)	1993	Paris (France)	23	12	52%
Morris (88)	1994	Salt Lake City (USA)	21	7	33%
Guinard (122)	1997	Paris (France)	10	4	40%
Gattinoni, Pesenti	2008	Milan, Monza (Italy)	124	49	40%
Total			268	116	43%

Table 21-4: Outcome with ECCO₂R in Milan-Monza (1979 to 2008; Adult Patients)

	Survivors	Nonsurvivors	p
No. Patients	49 (40%)	75 (60%)	
Age	33.9 ± 14.4	34.6 ± 15.2	NS
Days from intubation	11.9 ± 13.5	11.6 ± 9.1	NS
PaO ₂ /FI _{O2}	92.9 ± 44.6	76.3 ± 40.2	0.0353
Qs/Qt	0.46 ± 0.12	0.51 ± 0.10	0.0179
PaCO ₂ mm Hg	54.8 ± 17.2	63.9 ± 20.2	0.0106
PEEP cm H ₂ O	11.8 ± 5.5	13.2 ± 4.2	NS
PIP cm H ₂ O	41.8 ± 8.0	45.9 ± 10.5	0.0279
Cardiac index (L/min)	5.3 ± 1.6	4.9 ± 1.3	NS
Heart rate (bpm)	132 ± 19	125 ± 20	NS
BP mm Hg	84.5 ± 14.1	80.9 ± 13.7	NS
CVP cm H ₂ O	11.1 ± 5.7	11.3 ± 5.1	NS
PAP mm Hg	33.1 ± 8.9	35.3 ± 7.9	NS
WP mm Hg	13.1 ± 6.5	13.9 ± 5.0	NS
On vasopressor	6 (12%)	25 (32%)	0.0174
Days of ECCO ₂ R	15.7 ± 21.9	15.7 ± 16.3	NS

BP, arterial blood pressure, mean; CVP, central venous pressure; PAP, pulmonary artery pressure, mean; PIP, peak inspiratory pressure; Qs/Qt, intrapulmonary shunt; WP, pulmonary artery wedge pressure.

Only one controlled, randomized trial has been conducted on the effect of LFPPV-ECCO₂R in patients with severe ARDS.⁸⁸ The investigators enrolled forty patients meeting the original NIH-ECMO entry criteria. Nineteen patients were randomized to LFPPV-ECCO₂R and twenty-one to control mechanical ventilation. Survival was equivalent in the two groups: seven survivors in the ECCO₂R and eight in the control group. The investigators and accompanying editorialists⁸⁹ concluded that extracorporeal support is not recommended in ARDS.

A more balanced interpretation of this study, however, must take into account several considerations.^{90–92} The incidence of uncontrollable bleeding, leading to premature interruption of the treatment in seven of nineteen patients, was extremely high. More surprisingly, of seven ECCO₂R survivors, five had been disconnected as an emergency because of severe bleeding. Average transfused blood products (packed red cells plus fresh-frozen plasma) was 3.39 L/day (4.79 L/day in the survivors). A problem in the management of blood clotting or the surgical procedure to control bleeding appears obvious when related to contemporary published experience. This study suggests that despite a high rate of catastrophic bleeding in the ECCO₂R group, the net outcome was not worse in the ECCO₂R group. As such, ECCO₂R may prove beneficial, provided that the associated bleeding problems are handled effectively.

Recent Clinical Developments: Arteriovenous Bypass

Venovenous ECCO₂R still remains a complex procedure, reserved for centers with experience and capabilities to run it safely in very diseased patients. In an effort to simplify extracorporeal respiratory assist, Barthelemy et al⁹³ reported that an animal could be supported for up to 24 hours by a pumpless artery-to-vein extracorporeal system in combination with apneic oxygenation. Subsequently, Awad et al⁹⁴ demonstrated the feasibility of arteriovenous CO₂ removal (AVCO₂R) for up to 7 days in sheep. Young et al⁹⁵ evaluated AVCO₂R in a femorofemoral arteriovenous (AV) model, both with and without a blood pump. A step forward in pumpless AVCO₂R came with the design of very-low-resistance membrane lungs.⁹⁶ A pumpless system is expected to minimize the foreign-surface interactions and blood-element shear stress.

AVCO₂R (also termed *interventional lung assist* [iLA] or *pumpless extracorporeal lung assist*) was studied in normal animals and in experimental lung-injury models.^{97–100} Short-term feasibility and safety phase I trials were performed successfully in patients,¹⁰¹ showing that an AV shunt coupled with a low-resistance membrane lung can achieve an ECCO₂R between 70% and 100% of the total CO₂ production. Reng et al¹⁰² published a collection of ten patients treated by what they named “pumpless ECLA.” In 2006, a paper from the same institution reported ninety patients with ARDS treated with AVCO₂R, with a survival rate of 41%.¹⁰³ The authors pointed out that the contribution of the bypass to oxygenation was moderate, while the removal of CO₂ averaged 140 mL/min allowing a consistent decrease in ventilation. The paper also reported a high rate of complications, up to 24%, mainly related to the arterial cannulation. More recently Zimmerman et al¹⁰⁴ reported a higher survival rate (50.9%) and fewer complications (11%) with a reduction of arterial cannula size, allowing a residual lumen equal to or greater than 30% of the vessel diameter. The iLA Registry,¹⁰⁵ updated to 500 cases by July 2010, mostly ARDS (268 cases, 53.6%) and chronic obstructive pulmonary disease exacerbations (125 cases, 25%), points out that early institution of iLA produces a 42% higher survival rate. We are now awaiting the results from the prospective randomized Extrapulmonary Interventional Ventilatory Support for Lung Protection in Severe Acute Respiratory Distress (Xtravent) trial that completed enrollment on January 2011.¹⁰⁶

AV bypass can be established quickly, it is simple and does not require specialized personnel. The clinical application is limited to patients whose cardiovascular system can tolerate the increased cardiac output and whose arterial blood pressure can drive enough blood to achieve a sufficiently high CO₂ removal (shunt flows are 1 to 2.5 L/min). Although vasopressors can be added to increase shunt blood flow, no direct intervention is possible to otherwise regulate it. Lastly, AVCO₂R consists of pure CO₂ removal: with the exception of extremely severe hypoxemia, the amount of oxygen that can be transferred to arterial blood can influence the systemic oxygenation only marginally.¹⁰⁷ In contrast to venovenous ECCO₂R, no simple conversion to ECMO is possible. The technique is fascinating and promising in its extreme simplicity, however. Very-low-resistance, surface-heparinized membrane lungs are now available for clinical use.⁹⁷

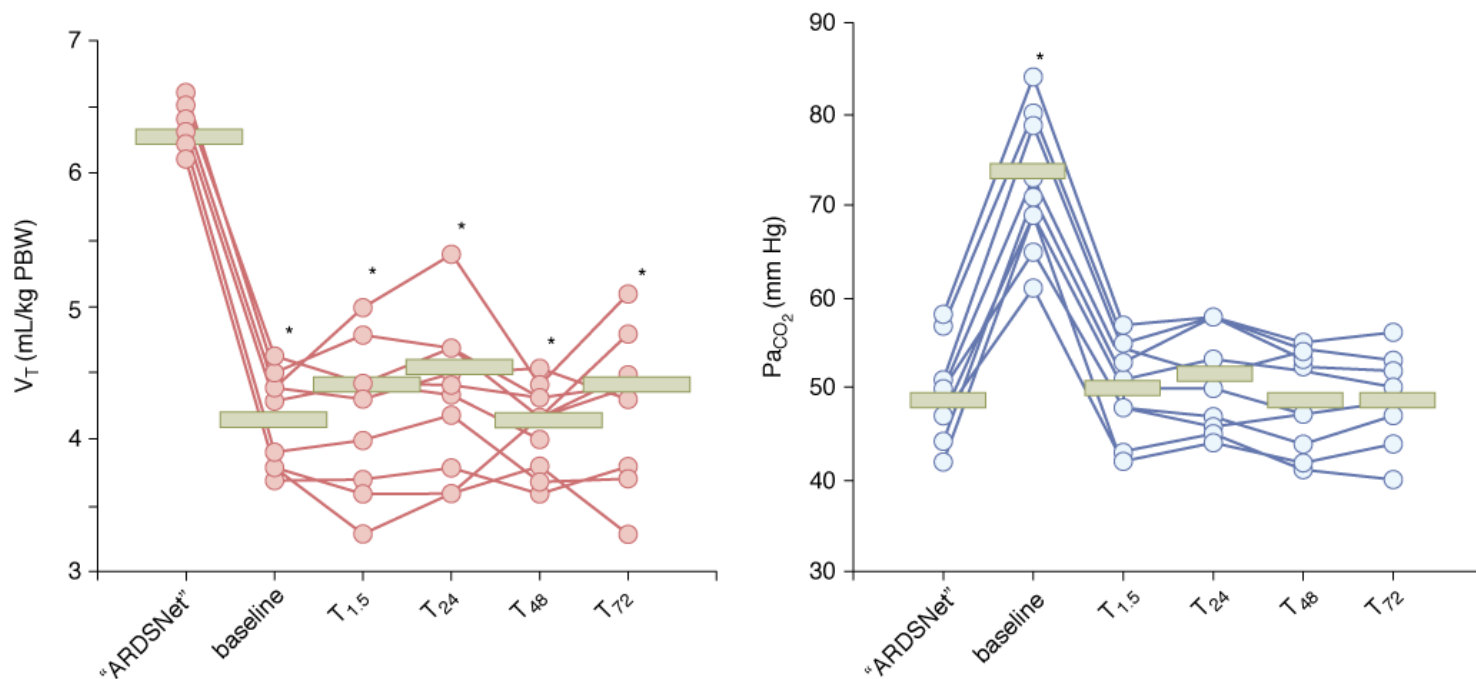
Pumpless devices have been recently used as bridge to lung transplantation^{108–110} with interesting results. The rapid implementation and extreme simplicity of the iLA system makes it suitable for the aerial transportation of critically ill patients with respiratory failure.^{111,112}

New Trends: Low Flow Partial Carbon Dioxide Removal

The clinical application of PECO₂R, originally described by Marcolin et al,⁴⁰ was successfully implemented in a patient with severe and persisting bilateral bronchopleural fistulas.⁹ The technique consisted of a very low flow (0.4 to 0.6 L/min) venovenous bypass, allowing a rapid liberation from mechanical ventilation to spontaneous breathing and healing of the fistulas.

More recently, Livigni¹ et al¹³ described the application of a modified continuous venovenous hemofiltration circuit comprising ultrafiltrate recirculation and a membrane lung to achieve partial CO₂ removal in sheep. In 2009, Terragni et al¹¹⁴ reported on the successful application of the same device to decrease tidal volume to levels lower than the standard 6 mL/kg predicted body weight (Fig. 21-6) in patients suffering lung hyperinflation in spite of lung-protective ventilation. The same investigator group¹¹⁵ reported that at least 30% of patients with ARDS undergo lung hyperinflation despite limiting tidal volume to 6 mL/kg.

Figure 21-6



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Individual and average (horizontal bar) values of tidal volume (V_T) and PaCO₂ during Acute Respiratory Distress Syndrome Network ("ARDSNet") strategy, after lowering V_T and before initiating carbon dioxide removal (baseline), and 60 to 90 minutes (T_{1.5}), 24 hours (T₂₄), 48 hours (T₄₈), and 72 hours (T₇₂) after initiation of carbon dioxide removal. * P versus ARDSNet strategy. The data show how CO₂ removal is able to maintain normocapnia despite a reduced, noninjurious tidal volume. (Reproduced, with permission, from Terragni PP, Del Sorbo L, Mascia L, et al. Tidal volume lower than 6 ml/kg enhance lung protection: role of extracorporeal carbon dioxide removal. *Anesthesiology*. 2009;111:826–835.)

Batchinsky et al¹¹⁶ successfully applied a new device (Hemolung) in swine that was able to remove a substantial proportion (72 mL/min) of an animal's CO₂ production at blood flows comparable to those used in dialysis or hemofiltration (400 to 500 mL/min). Preliminary clinical experience in five spontaneously breathing patients with chronic obstructive pulmonary disease has been recently reported.¹¹⁷

Zanella et al¹¹⁸ achieved the remarkable result of a CO₂ removal of more than 150 mL/min from a blood flow of 500 mL/min in pigs. The technique includes the administration of a metabolizable acid (lactic acid) to convert blood bicarbonate to gaseous CO₂ before the membrane lung.

Conclusions

ECCO₂R is a fascinating approach to the management of respiratory failure and is a powerful tool for overcoming any ventilatory problem. For patients with the most severe form of ARDS, a venovenous circuit with the possibility of shifting to modern full venovenous ECMO (if needed) may be a better solution. Venovenous ECCO₂R should be limited to centers where appropriate technical skills, motivations, personnel, and experience are available.

In its purest conceptual application, ECCO₂R is achieved by an arteriovenous pumpless shunt. Exciting perspectives are offered by the development of low or very low blood flow extracorporeal CO₂ removal techniques.^{113–118} Their applications open up the possibility of substantially limiting the invasiveness of mechanical ventilation, to decrease the rate of failure of noninvasive ventilation, to limit the use of endotracheal intubation, and therefore substantially limit the incidence of ventilator associated pneumonia (actually an intubation-related disease). The need for increased airway pressure could be essentially covered by invasive or noninvasive continuous positive airway pressure, and sedation needs might as well be markedly decreased.

The dream of a fully awake, nonintubated, eating, drinking, and communicating patient connected to something like a hemodialysis machine, rather than a fully sedated, intubated, mechanically ventilated patient might finally have a chance of becoming true.^{119,120}

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